

SERVICE MANUAL



Document 2380634-4

Revision 1

**GE Signa® 3.0T
PREMIER III
4-CHANNEL CERVICAL-THORACIC-LUMBAR (CTL) ARRAY**

GE Catalog Part Number: M1385AB

USAI Part Number: 160200

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Rev. 08/15/2003

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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

To identify the 4-Channel CTL Array, refer to the coil label (see *Figure 1*).

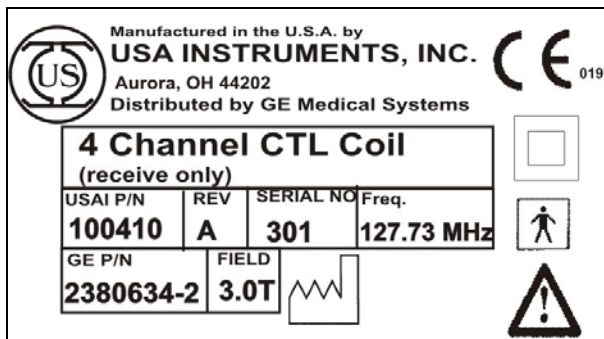


Figure 1: Coil label.

Figure 2 shows a picture of the 4-Channel CTL Array.



Figure 2: The 4-Channel CTL Array.

SHIPPING LIST – TABLE 1-1

Box #	Part Name	GE Part #	USAI Part #	Qty
1	Coil	2380634-2	100410	1
2	Phantom Kit	2377426-3	550011	1
3	Coil Pad	2380634-5	150504	1
3	Ear Pad	2380634-6	150503	2
3	Ear Pad Covers	2380634-8	150537	1
3	Knee Pad	2380634-9	150529	1
1	Operator's Guide	2380634-3	770053	1
1	Service Manual	2380634-4	780049	1

1-2 Compatibility

This coil is compatible with the GE Signa® 3.0T Eclipse system.

1-3 Related Documentation

Operator's Guide, GE Part Number 2380634-3.

Signa® 3.0T Eclipse Service Methods CD, GE Part Number 2360530.

1-4 Environmental Requirements

Storage Requirements

The coil should be stored in the scanner room.

Dimensions

43.3" L x 16.1" W x 9.3" H (110.0 cm x 41.0 cm x 23.5 cm)

Weight

20.0 lbs. (9.1 kg)

1-5 Theory of Operation

The 4-Channel CTL Array is a receive-only phased array coil with eight element coils. They are arranged into four sections in head-foot direction. Each section has two element coils. *Figure 3* shows the layout of element coils and sections on the coil former. Use Section 1 and 2 to scan the cervical spine, Sections 2 and 3 to scan the thoracic spine and Sections 3 and 4 to scan the lumbar spine. In each case the DC currents from the system turn off the element coils that are not selected. The eight element coils are directly connected to the eight pre-amplifiers of the system. *Figure 4* shows the connection between the element coils and the system interface. Each element coil is matched to 50Ω with appropriate loading in order to get the best SNR. The DC currents are input from the RF output cables to the element coils. To prevent coupling to the transmit B_1 field, each coil element in the coil has active and passive decoupling circuits. *Figure 5* illustrates the active and passive decoupling circuits in each element. The coil diagram is shown in *Figure 6*.

The active blocking circuit is switched on and off through DC bias on PIN diode. During the transmission period, the PIN diode is turned on as shorted circuits, with a forward bias DC generated and controlled by the MRI system. The switching diodes in passive decoupling circuit are turned on by the induced RF voltage in the coil circuitry, coupled from the transmit field, and charge the bias feed capacitor. This capacitor provides a forward bias current to turn on the PIN diode in the passive decoupling circuit. In each decoupling circuit when the PIN diode is turned on the inductor resonates with the local coil capacitor. Because of the low resistance of the PIN diode each decoupling circuit has very high resonant impedance (typically above $2\text{ K}\Omega$), compared with the coil resistance (less than 50Ω). These high impedance elements in the coil circuit segregate the coil circuitry into several isolated electrical segments and deresonate the coil. Therefore, the deresonated coil does not support any RF current flow in its circuit that might be induced by the transmission coil.

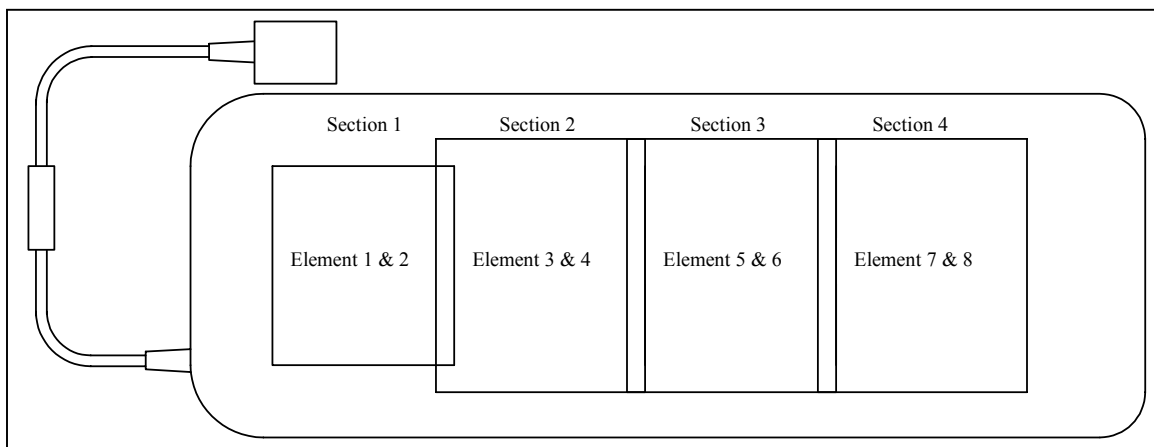


Figure 3: Coil physical layout.

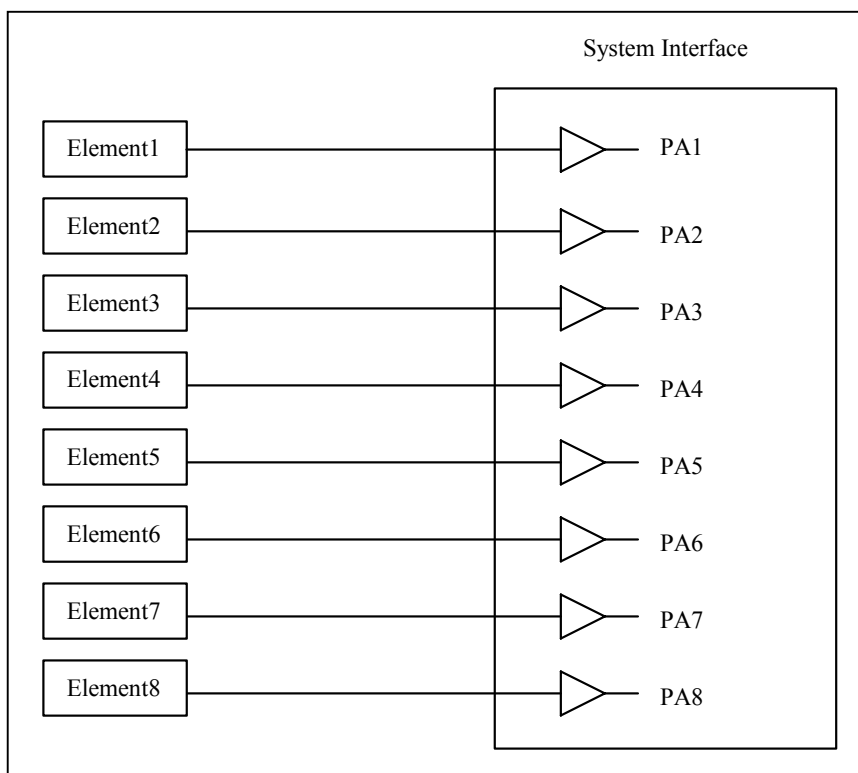


Figure 4: Element coils and system interface connection.

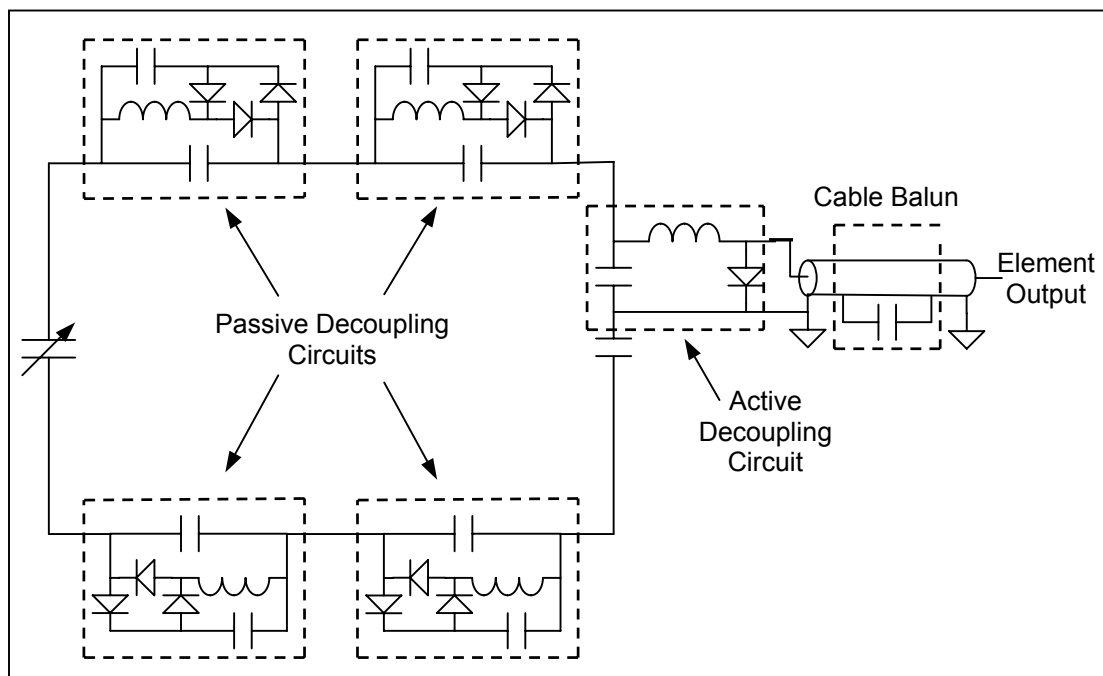


Figure 5: Decoupling circuits of the element coil.

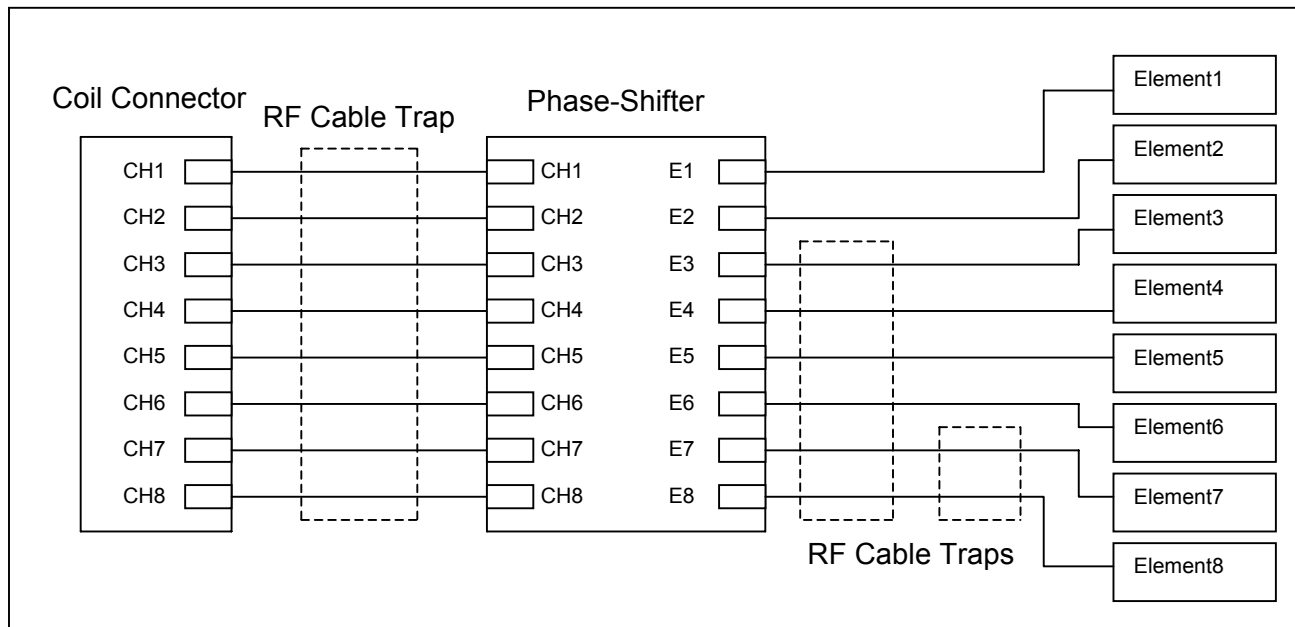


Figure 6: Diagram of the coil.

SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None.

2-1-2 Configuration

No configuration should be required. For reference, the configuration information is included in the Appendix; however, that configuration is only current as of the printing of this manual.

2-2 Installation Functional Checks

1. From the Scan Desktop, start new scan by selecting [**New Pt**]; set **Patient ID** to “geservice” and **Patient Weight** to “111” pounds. Click [**Patient Position**] to open protocols window.
2. Plug the coil into the patient table.
3. Perform Section 3-2 Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification and record data values in Data Sheet.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	USAI Part #	Qty
Phantom Kit	2377426-3	550011	1

3-2-2 Explanation of Procedure

To fully test SNR on the 4-Channel CTL Array, two sections will be scanned. These sections are identified as 4CTL12 and 4CTL34. SNR measurements must be made for each section, requiring two sets of signal and noise scans in the coronal plane.

Refer to the Data Sheet in Section 3-2-6 to understand the data required to calculate the composite SNR for each section of the coil.

The image quality check uses two different protocols for signal and noise image acquisition. The signal scan is an **FSE** sequence used to minimize susceptibility and B_0 inhomogeneity effects. The noise scan is a **GRE** sequence that has a Control Variable (do_noise) to eliminate the transmit RF completely during the scan. The signal scan **must** be run prior to the noise scan as the R1, R2, and TG values from the signal scan are used for the noise scan.

3-2-3 Signal Scan

1. From the **Scan Desktop**, start new scan by selecting [**New Pt**]; set **Patient ID** to “geservice” and Patient Weight to “111” pounds. Click [**Patient Position**] to open protocols window.
2. Position the phantom set on the coil without the patient pad (see *Figure 7*).
3. Landmark on the vertical line corresponding to the region of interest (see *Figure 7*).

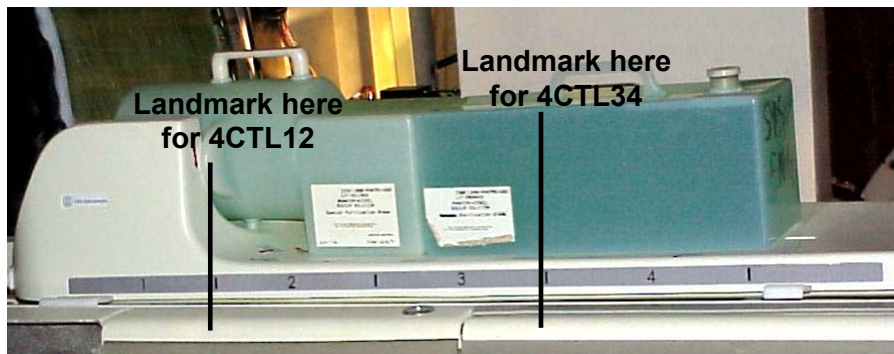


Figure 7: Landmarking for signal and noise scans.

4. At the magnet, press “**Alignment Light**” to turn on the light. Move the cradle to align the coil to the alignment lights. Press “**Landmark**” to landmark the alignment.
5. Move the coil to scan position by clicking on the “Move to Scan” button.
6. At the console, verify the coil has been properly identified by the system: correct picture on the screen and name in the coil field. If the system does not recognize the coil, refer to Section 2 - Setup and Calibration in the Service Manual.
7. Set the protocols per the Signal section from Table 3-2-4 Signal and Noise Protocols.
8. Select [**Save Series**] the [**Prepare to Scan**].
9. Prior to the scan, open the [**Display CVs**] menu under [**Research Operations**].
10. Set the “**saveinter**” CV to 1, saving the intermediate images. This is required to ensure all channels are operating well.
11. Select [**Auto Prescan**].
12. Record the R1, R2 and TG values obtained, then select [**Scan**].
13. If not continuing to noise scan, be sure to end the exam when finished. This will ensure that all CVs return to their default values.

3-2-4 Noise Scan

A signal scan must be run prior to the noise scan as the same R1, R2 and TG values must be used for both the signal and noise scans. Do not run an Auto Prescan prior to the noise scan as the values will be changed.

1. Copy the signal scan series. Use [**Copy Series**] (highlight signal series and click right mouse button) and [**Paste Series**] in **RX Manager**.
2. Click [**View Edit**] and set the protocols per the Noise section from Table 3-2-4 Signal and Noise Protocols.
3. Click [**Save Series**] and click [**Prepare to Scan**].
4. Open the [**Display CVs**] menu under [**Research Operations**].
5. To obtain the noise images, set the “**rhformat**” and “**do_noise**” CVs to 1.
6. Start [**Manual Prescan**], look at the power spectrum and verify R1, R2, TG and Bandwidth have not changed from the previously completed signal scan.
7. Select [**Done**] and exit without changing any parameters.
8. Select [**Scan**].

SIGNAL AND NOISE PROTOCOLS – TABLE 3-2-4

	Signal	Noise
Patient/Exam Information		
Patient ID	geservice	geservice
Patient Name	CTL Array	CTL Array
Patient Weight	111 lbs. (50 kg)	111 lbs. (50 kg)
Patient Position	Supine	Supine
Patient Entry	Head First	Head First
Coil	4CTL12/4CTI34	4CTL12/4CTI34
Series Description	Signal	Noise
Imaging Parameters		
Plane	Coronal	Coronal
Mode	2D	2D
Pulse Seq	FSE	GRE
Imaging Options	none	none
PSD Name	<i>leave blank</i>	<i>leave blank</i>
Protocol	<i>leave blank</i>	<i>leave blank</i>
Scan Timing		
# of Echoes	1	1
TE	17	minful
TR	500	34
Echo Train Length/Flip Angle	4	1
Bandwidth	15.63	15.63
Additional Parameters		
<i>no entries required in this area</i>		
Acquisition Timing		
Freq	256	256
Phase	256	256
NEX	1	1
Phase FOV	1.00	1.00
Preq DIR	S/I	S/I
Auto Center Freq	Peak	Peak
Autoshim	On	On
Phase Correct	On	On
Contrast	Off	Off
# of Reps B4 Pause	0	0
Scanning Range		
FOV	24	24
Slice Thickness	3	3
Spacing	1.5	1.5

	I/S Center	P/A Center	R/L Center
Start	0	0	
End	0		
# Slices	1	Table Delta	0.0

3-2-5 SNR Image Analysis

Image Analysis

Regions of interest in both signal and noise images can be measured directly in the image browser. Click the user interface button **Measure**, select the square or rectangular shape, and adjust its size and orientation when the shape is displayed in the selected image. Mean, standard deviation, and area of the ROI will appear in the lower right corner of the image.

Examples of typical intermediate Receiver Images for 4CTL12 are shown in *Figure 5*. Typical composite and noise images for 4CTL12 are shown in *Figures 6 and 7*.

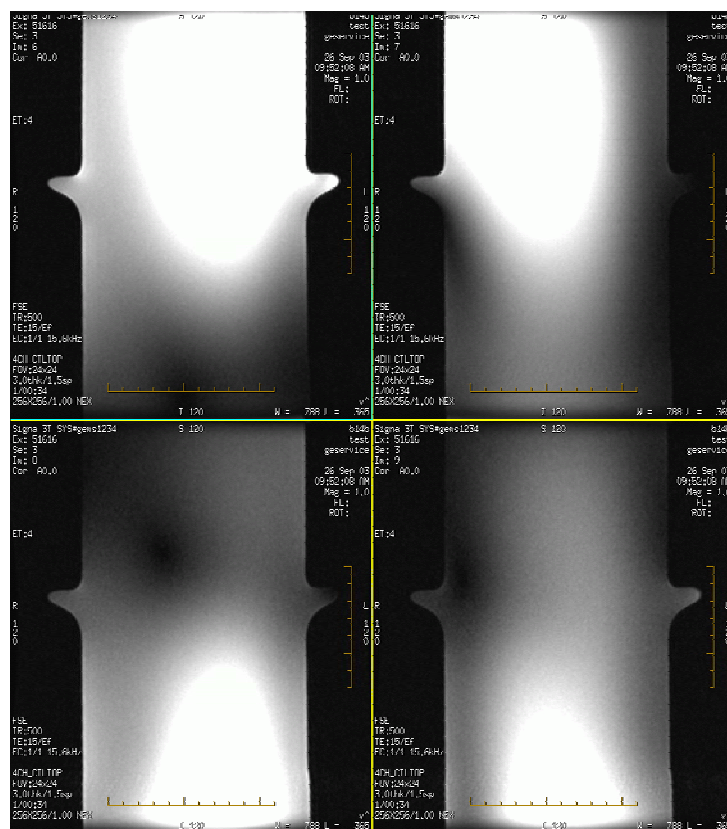


Figure 5: 4CTL12 intermediate receiver images 1-4.

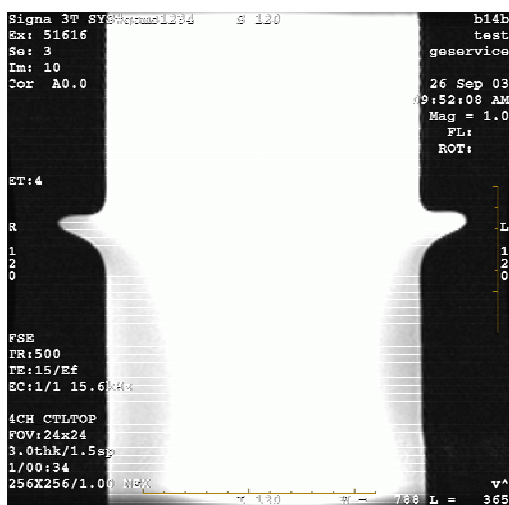


Figure 6: 4CTL12 composite image.



Figure 7: 4CTL12 noise image.

Examples of typical intermediate Receiver Images for 4CTL34 are shown in *Figure 8*. Typical composite and noise images for 4CTL34 are shown in *Figures 9 and 10*.

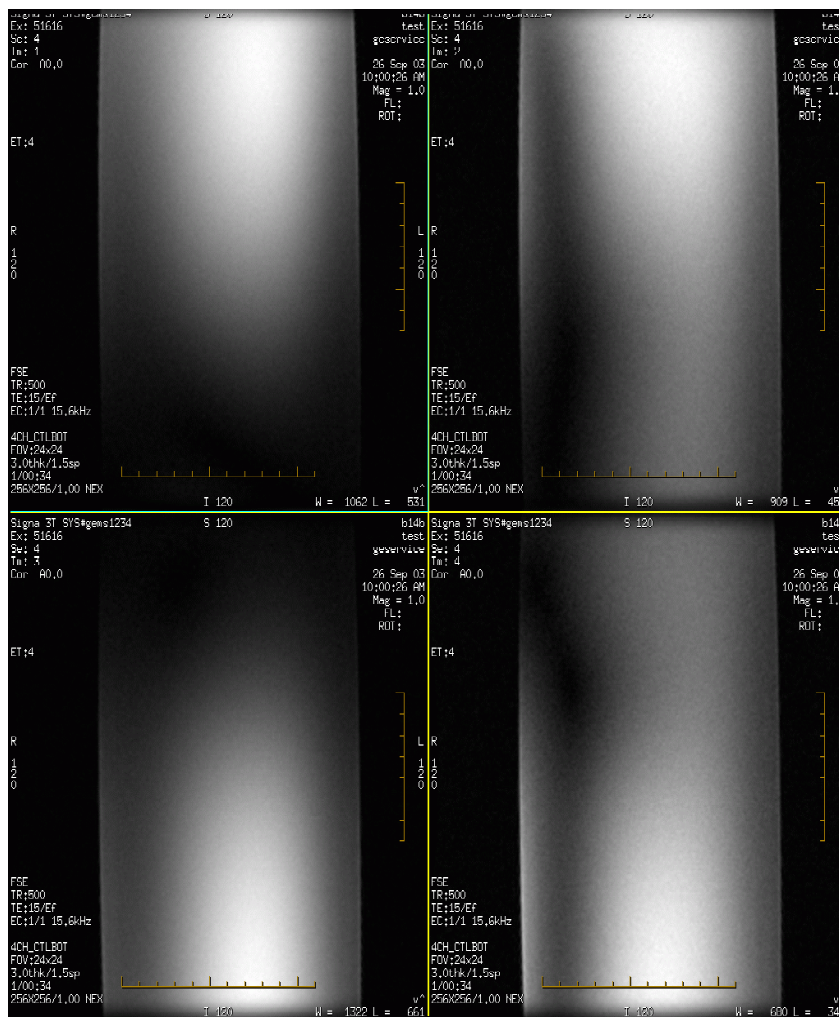


Figure 8: 4CTL34 intermediate receiver images 1-4.

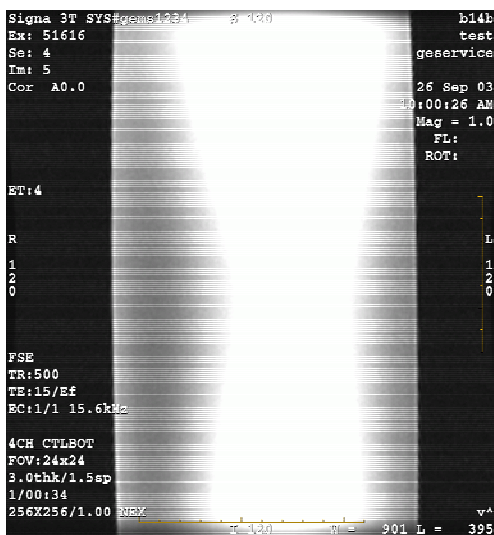


Figure 9: 4CTL34 composite image.

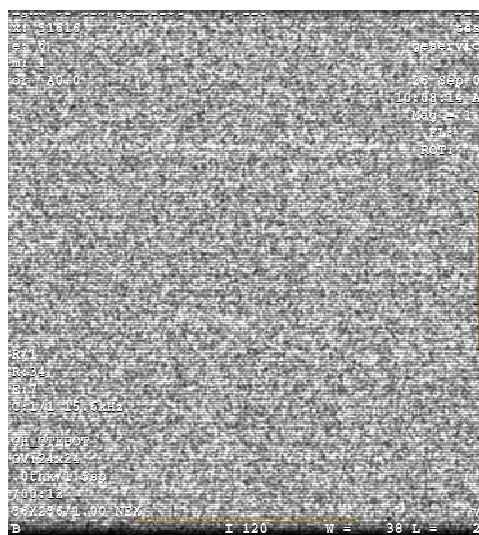


Figure 10: 4CTL34 noise image.

SNR Measurement

To acquire the composite signal image, take two SNR measurements for 4CTL12. A 10cm square ROI should be used for the top two set of elements in the cervical section of the coil (see *Figure 11*). For the Noise image, a 24cm square ROI should be used (see *Figure 12*).

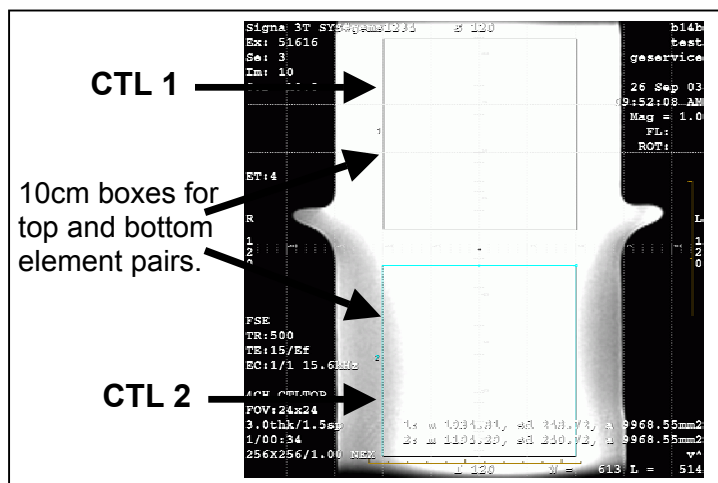


Figure 11: Composite 4CTL12 SNR ROIs.

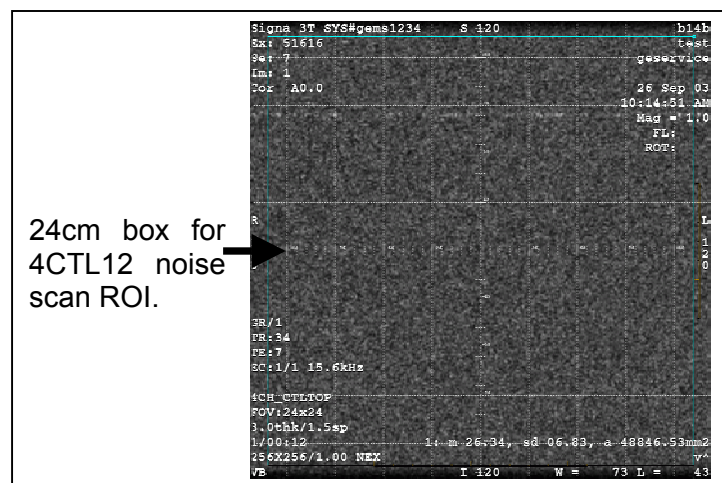


Figure 12: 4CTL12 noise ROI.

To acquire the composite signal image, take one SNR measurement for 4CTL34. A rectangular 20cm height and 10cm width ROI should be used for the bottom two sets of elements in the lumbar section of the coil (see *Figure 13*).

For the noise image, a 24cm square ROI should be used (see *Figure 14*).

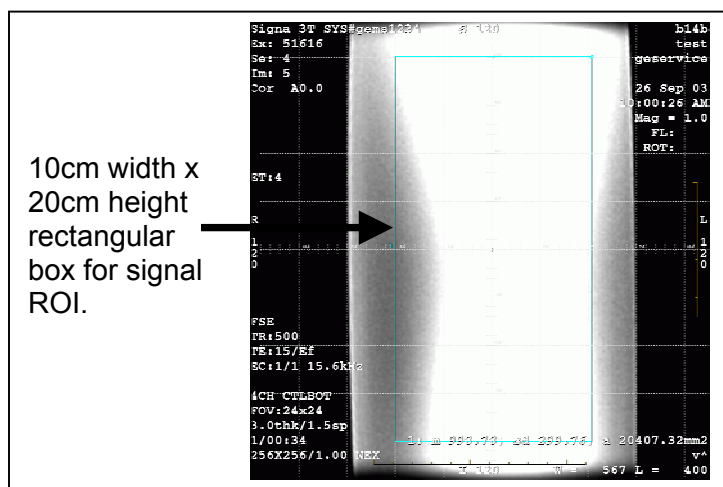


Figure 13: Composite 4CTL34 SNR ROI.

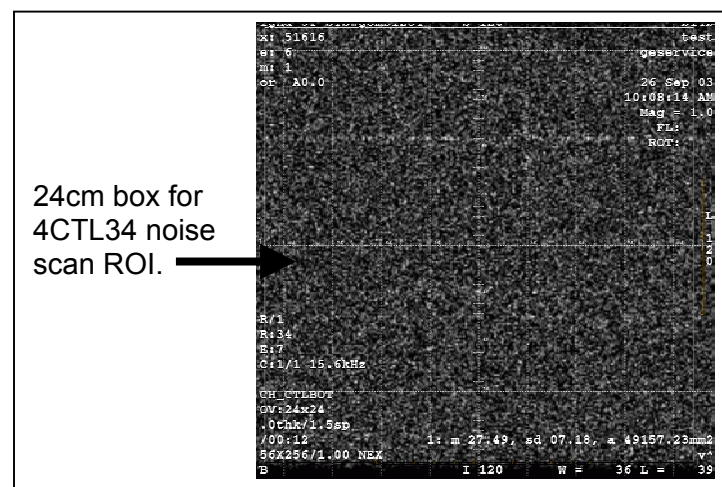


Figure 14: 4CTL34 noise ROI.

The SNR is calculated using the signal to noise ratios of the individual receiver channels:

$$\text{SNR}_{\text{Composite}} = \frac{\text{Mean of Signal within ROI}}{\text{Standard Deviation of Noise within ROI}}$$

Note: The SNR calculation uses the **MEAN** of the signal image and **STANDARD DEVIATION** of the noise image. SNR is measured on the composite image.

SNR Specification

The SNR measurements must be greater than or equal to the following specifications:

SNR SPECIFICATIONS – TABLE 3-2-5

Coil Element	Composite SNR
CTL1	240
CTL2	145
CTL34	117

3-3 External Cable Check

Check for continuity in the output cable.

Step 1 – Open the coil bottom cover. Remove the cable clamp. Disconnect the RF cables with the phase-shifter board (located near the top of the coil). Remove the cable from the coil.

Step 2 – Refer to *Figure 15* and Phased Array Connector Pin Assignments – Table 3-3 for the correct pins. Use a Digital Volt Meter (DVM) to check continuity between the appropriate pins at the 30-pin Bendix connector and the RF cable connectors that connect to the phase-shifter board.

CENTER PIN AND GROUND CONNECTIONS – TABLE 3-3

System	Pin:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Eclipse																
Row A		G	NC	G	NC	G	NC	G	NC	G	G	G	NC	G	NC	G
Row B		S1	NC	S2	NC	S3	NC	S4	NC	S5	G	S6	NC	S7	NC	S8

(G = Ground, S1 = RF Signal Input for Channel 1, NC = No Connection, etc.)

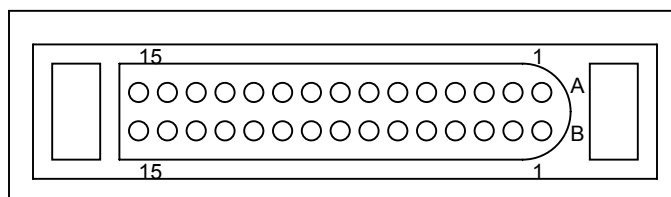


Figure 15: Pin layouts of 30-pin Bendix connector.

3-4 PIN Diodes Check

Step 1 – Remove the coil bottom cover.

Step 2 – Check the two active decoupling PIN diodes located on each of the feeding boards with a DVM (see *Figures 16 and 17*).

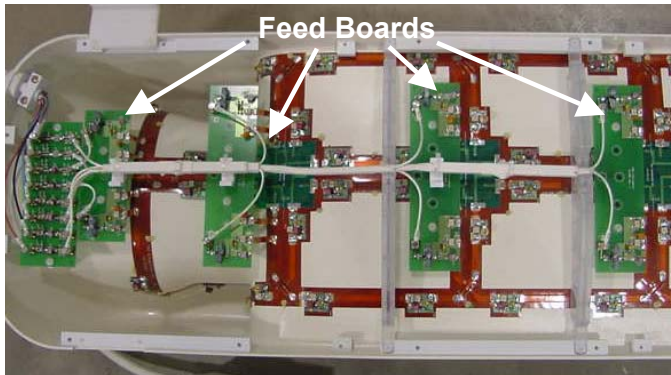


Figure 16: Location of the feeding boards on the coil.

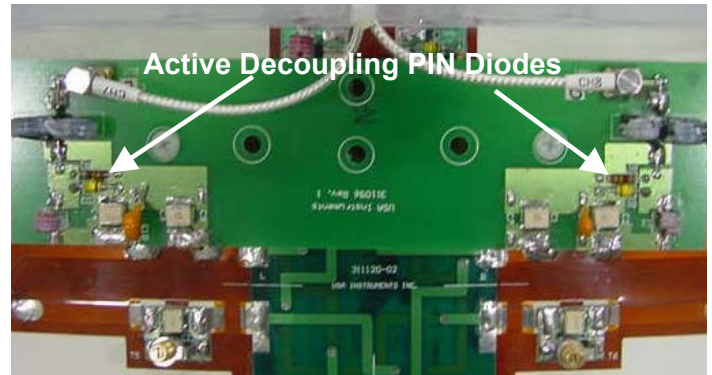


Figure 17: Location of the PIN diodes on the feeding board.

Step 3 – Check the passive PIN diode and two signal diodes located on the passive decoupling board with a DVM (see *Figure 18*).

Step 4 – If a diode is found to be either open or shorted:

- Take note of the diode's cathode-anode orientation.
- Carefully de-solder the diode from the PCB board.
- Cut the leads of a new diode to an appropriate length.
- Solder the diode to the original pads on the PCB.

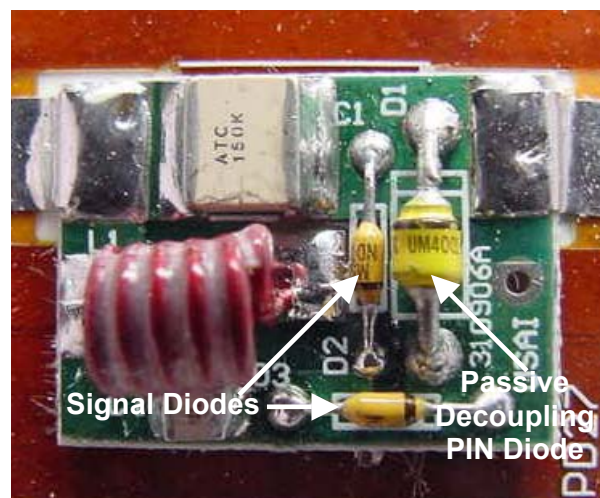


Figure 18: Location of the pin diode and signal diodes on the passive decoupling board.

3-5 Mechanical Hardware Check

None.

3-6 Troubleshooting Tips

When troubleshooting the coil, refer to chapter 3-2 Coil Imaging Performance Verification for imaging scan.

Symptom #1: The system reports a coil fault during prescan.

Probable Cause	Suggested Actions	Resolution
The coil connector has become disconnected from the system interface.	Check to make sure the coil connector is fully engaged.	Engage connector and try the scan again.
There is a DC short or open somewhere within the DC path inside the coil.	Refer to the Chapter 3-3 External Cable Check for pin layout of 30-pin Bendix connector. Turn DVM to Diode Check mode and check the signal pins to the ground. Try moving and twisting the cable as you watch the meter to find any intermittent connections.	Step 1- If a short or open is observed, remove the coil bottom cover. Step 2- Check the active PIN diodes (refer to Chapter 3-4 Diodes Check). If a short or open is observed, change the PIN diode. Step 3 - If no PIN diode is damaged, check the coil cable (refer to Chapter 3-3 External Cable Check). If the coil cable has no problem, replace the coil. Otherwise replace the coil cable.
One or more active decoupling PIN diodes have failed.	Referring to Chapter 3-4 Diodes Check, check the active decoupling PIN diodes using a DVM.	Replace the PIN diode.
The output cable has a short or an open.	Referring to Chapter 3-3 External Cable Check, check the coil cable a DVM.	Replace the coil cable.
There is a problem with the DC bias from the system interface	Check to ensure the correct DC bias is being provided to the coil during transmit and receive per GEMS System Service Manual. Try a similar 4-channel coil to see if the same problem occurs	Correct MRI system problem.

Symptom #2: The coil does not pass SNR tests or exhibits poor image quality on patient scans.

Probable Cause	Suggested Actions	Resolution
One or more of the coil channels has a high noise level.	Look at the uncombined images to determine which channel has the problem. Compare the noise standard deviation measurements between channels and system performance logs; they should be approximately the same.	Replace the coil.
One or more of the coil channels has low signal.	Look at the uncombined images to determine which channel has the problem. Compare signal mean measurements between channels and system performance logs.	Replace the coil.
Excessive ghosting is causing the noise standard deviation measurements to be artificially high.	Window and level the images down to look at the background for signs of ghosting. Try padding the phantom in the holder. Try running the scan without the phantom holder to see if the ghosting diminishes. Try turning off the cold heads to see if the ghosting is vibration-related.	If padding or changing the phantom position can minimize ghosting, then the problem is caused by excessive vibration elsewhere in the system. If the ghosting is not symptomatic of phantom positioning or padding, then replace the coil.
There is a problem with the DC bias from the system interface.	Check to ensure the correct DC bias is being provided to the coils during transmit and receive per GEMS System Service Manual.	Correct MRI system DC bias problem.

SECTION 4 – MAINTENANCE

4-1 Coil Care



WARNING!

Detach the coil connector from the system before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning the coil until the coil has dried completely. Electric shock may result if the coil is attached to the system during cleaning or when it is wet.



CAUTION

Do not spray or pour cleaning solution directly on the coil. Do not submerge the coil in the solution. The coil contains sensitive electronic components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: (1) a ten percent bleach solution (some discoloration may occur), (2) one ounce commercial dishwashing liquid mixed with one gallon of water or (3) warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning a coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

4-3 Carrying the Coil

The coil should be supported from underneath using both hands. Do not carry the coil by the cable.

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 External Cable Replacement

Step 1 – Remove coil cover.

Step 2 – Remove the two screws on the cable clamp that hold down the cable assembly.

Step 3 – Disconnect SMB plugs on the cable end from the phase-shifter boards.

Step 4 – Remove the old cable assembly.

Step 5 – Fix a new cable assembly on the CTL coil and reattach the cable clamp with two screws. Connect all eight SMB plugs to the corresponding receptacles on the phase-shifter board (see *Figure 20*).

Step 6 – Close the coil cover.

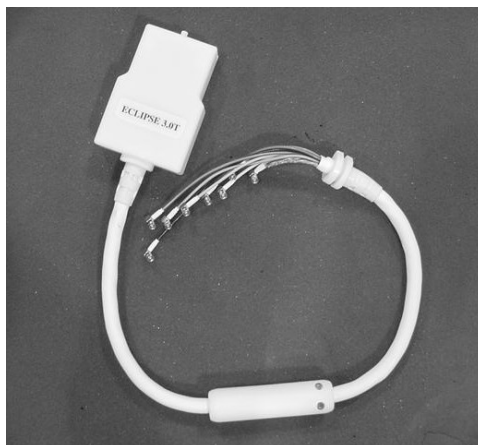


Figure 19: External cable assembly.

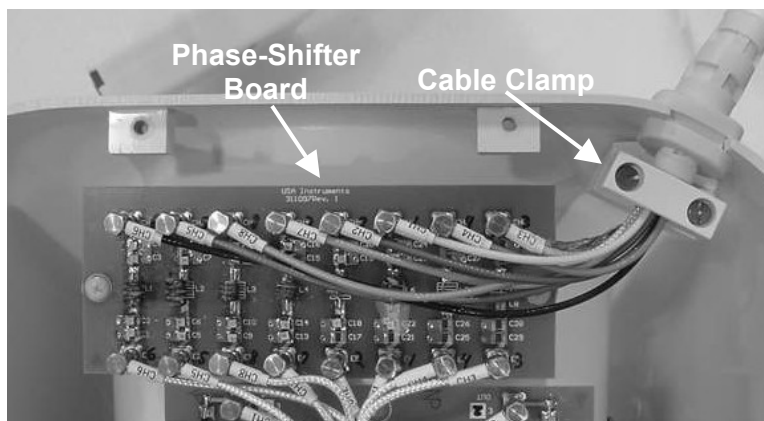


Figure 20: Fixing the cable assembly on the phase-shifter board.

5-2 Mechanical Hardware Replacement

None.

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS – TABLE 6-1

Part Name	GE Part #	USAI Part #
Coil	2380634-2	165155
Cable Assembly	2380634-7	155217
Phantom Kit	2377426-3	155218

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES – TABLE 6-2

Part Name	GE Part #	USAI Part #
Coil Pad	2380634-5	155213
Ear Pad	2380634-6	155214
Ear Pad Covers	2380634-8	155216
Knee Pad	2380634-9	155215

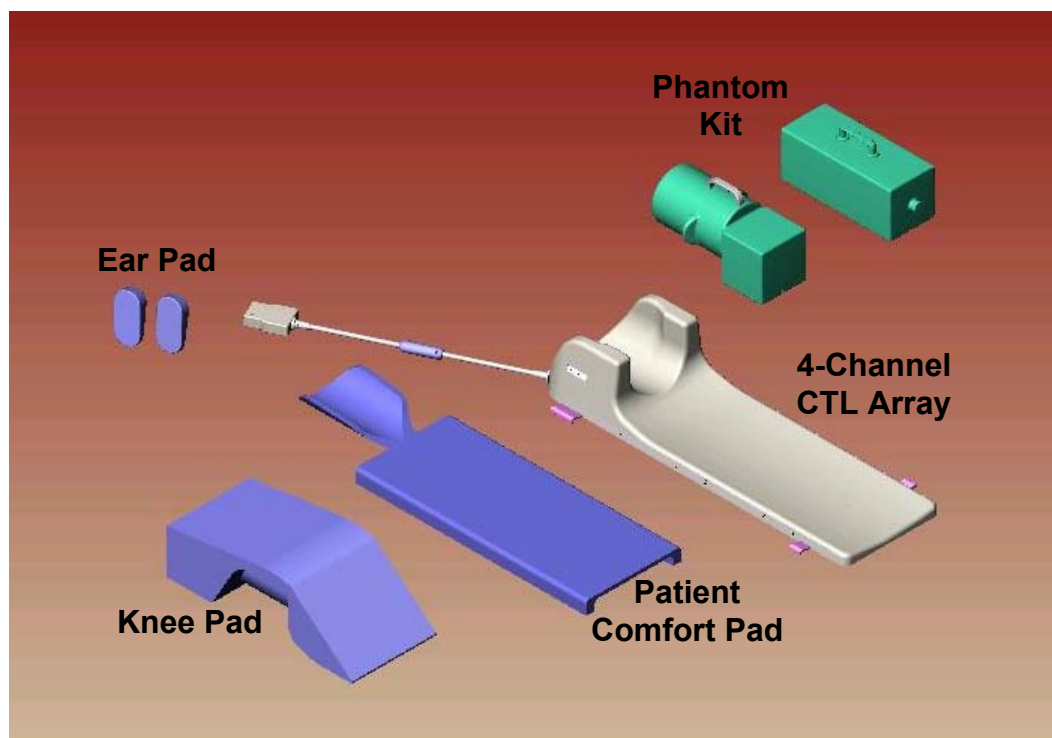


Figure 19: Parts of the coil.

SECTION 7 – APPENDIX

7-1 SNR Data Sheet

Use the table provided below to record the calculated signal to noise ratio (SNR) data obtained from the Functional Checks section.

Date	Comments						
Mode	R1	R2	TG	Composite Image Signal Mean	Noise Std Dev	SNR	Spec Limit
4CTL1							
4CTL2							
4CTL34							

Date	Comments						
Mode	R1	R2	TG	Composite Image Signal Mean	Noise Std Dev	SNR	Spec Limit
4CTL1							
4CTL2							
4CTL34							

7-2 Coil Configuration

Parameter	Modes				
Coil Name	4CTL12	4CTL23	4CTL34	4CTL1	4CTL4
Coil Type	3	3	3	3	3
Extremity Coil	no	no	no	no	no
Cable Loss	1.05	1.05	1.05	1.05	1.05
Coil Loss	0.975	0.975	0.975	0.975	0.975
Recon Scale Factor	1.00	1.00	1.00	1.00	1.00
Linear vs. Quadrature	1	1	1	1	1
Multiple Receiver Coil?	yes	yes	yes	yes	yes
Number of Receivers	4	4	4	2	2
Starting Receiver ID	0	0	0	0	2
Ending Receiver ID	3	3	3	1	3
Multi-Coil Port Enable	3	6	12	1	8
Multi-Coil Port Error Enable	3	6	12	1	8
Additional Transmit Attenuation	0	0	0	0	0
Number of Fast Receivers	1	1	1	1	1
Starting Fast Receiver ID	4	4	4	4	4
Ending Fast Receiver ID	4	4	4	4	4
Start TA Value	90	90	90	90	90
Start RG Value	12	12	12	12	12
Phased Array T/R Coil for Autoshim	-1	-1	-1	-1	-1
Multi-Coil Recon Enable	0	0	0	0	0
Head Default Freq Direction	0	0	0	0	0
SCIC Axial	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100
SCIC Sagittal	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100
SCIC Coronal	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100
Body Percent Exposed	100	100	100	100	100
aps1_mod	0	0	0	0	0
aps1_plane	1	1	1	1	1
aps1_mod_fov	200	200	200	200	200
opthickPS_mod	5	5	5	5	5
aps1_mod_PStloc	0	0	0	0	0
aps1_mod_PSrloc	0	0	0	0	0
Coil Transmit Position	2	3	3	2	3

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
1	11/03	Teresa DeMarco	First Issue